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A system dynamics model for emissions projection of hinterland transportation

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ABSTRACT

The rapid development of China's economy has led to a rapid increase in port throughput and made the issue of hinterland transport emissions more prominent. In this paper, a system dynamics model was established to project the long-term emissions of hinterland transportation. The model functions in a modular manner, consisting of core subsystems developed based on the classic four-step model of transportation, and several additional subsystems that are used to address the characteristics of the hinterland transport processes. The developed model was applied to a specific case in China to assess the long-term effects of alternative environmental policies on hinterland transport emissions. Through a comprehensive comparison of different policy scenarios, the model demonstrates explicitly how alternative environmental policies affect long-term emissions through a systematic intervention on a series of hinterland transport processes. The model also identifies the effective policies by which a lower emissions stock could be achieved. In addition, some suggestions for the policy making in hinterland transportation are provided.

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1. Introduction

In 2008, China overtook the United States to emerge as the world's largest emitter of carbon dioxide. Freight transportation sector is a major contributor to carbon and atmospheric pollutant (AP) emissions. Hinterland transportation, as an important component of national freight transportation systems, refers to freight transport between a port and an inland destination (De Langen et al., 2013). Dry bulk, general cargo, and containers constitute a large portion of the freight volume of hinterland transport systems in China. The typical modes of transportation are land (through highways and railways) and water (through inland waterways for ports along rivers). In the past two decades, the throughput of primary seaports in China has increased remarkably owing to the rapid development of the nation's economy, growing from 0.8 billion tons in 1995 to 8.1 billion tons in 2016 (National Bureau of Statistics of China, 2017). With the increase in port throughput, the issue of hinterland transport emissions has become more prominent.

Hence, several departments of the Chinese government, including the National Development and Reform Commission, and the Ministry of Transport, have issued a series of environmental policies to encourage the frequent use of more efficient transport modes to reduce hinterland transport emissions (National Development and Reform Commission, 2016; Ministry of Transport of China, 2016). Compared to trucking, transport by trains and barges presents advantages such as lower environmental strain, lighter port city traffic, lower transport distance costs, and faster throughput in ports (Woxenius and Bergqvist, 2011). These proposed environmental policies are in nature modal shift polices that shift freight away from highways using fiscal measures (e.g., rail freight funding or highway freight taxation), regulatory measures (e.g., regulation on truck weight and compulsory adoption of green vehicles), and infrastructure construction (e.g., constructing new rail infrastructures) (Bickford et al., 2014).

From the perspective of policy enforcement, these environmental policies that require new construction of freight infrastructures and promotion of green freight vehicles, often incur high expenditures. Meanwhile, hinterland transportation processes constitute a complex system in which multiple components (e.g.,

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Abbreviations

AP	Atmospheric Pollutants
CM	Corrective Maintenance
DN	Do Nothing
FGD	Freight Generation and Distribution
H2P	Hinterland-to-Port
HID	Hinterland Infrastructures Dynamics
HTE	Hinterland Transport Emissions
HTT	Hinterland Transport Time
LNG	Liquefied Natural Gas
LTR	LNG Trucks Replacement
MFE	Modal Fleet Evolution
MSF	Modal Split of Freight
MSR	Modal Shift from highways towards Railway
P2H	Port-to-Hinterland
PCU	Passenger Car Units
PPI	Pavement Performance Indicator
RM	Restorative Maintenance
RPL	RWR Plus LTR
RPM	RWR Plus MSR
RWR	Rigid Weight Regulation
SD	System Dynamics

cargo types, transport modes, infrastructure capacity, transport cost, and transport time) are interlinked (Halim et al., 2016). Changes established by the polices in one component are thus transmitted to other components and can be fed back to the original source of variation, causing far-reaching impacts to hinterland transport emissions in the long term. In this regard, decision-makers should emphasize on policy enforcement, as improper decisions are not only confronted with risks of invalid investment, but can also aggravate the original problems and increase emissions. Therefore, an ex-ante analysis of the effects of these environmental policies on hinterland transport emissions in the long term is meaningful, and can provide insights for decision-makers.

The objective of this study is to develop a system dynamics (SD) model for emissions projection of hinterland transportation. The SD model evaluates the long-term effects of alternative environmental policies on hinterland transport emissions based on the simulation of a series of related hinterland transport processes, and identifies those effective policies through which a lower emissions stock could be achieved.

2. Literature review

Research on hinterland transportation has been conducted since the 1990s. Previously, van Klink and van den Berg (1998) defined hinterland as the continental areas of origin and destination of traffic flows through a port. Hinterland transportation, correspondingly, is the transport from the port to the final destination, and vice versa (De Langen et al., 2013). The issue of hinterland transport emissions has thereafter attracted increasing attention as port throughput increases with globalization. For example, some researches investigated the potential of emissions reduction through the optimization of hinterland transport processes (Iannone, 2012a, 2012b; Roso, 2007); and some evaluated the effects of modal shift measures to the emissions reduction of hinterland transportation (Bergqvist et al., 2015; Jensen and Bergqvist, 2013; Woodburn, 2013). However, existing studies regarding hinterland transport systems in China still contain some limitations.

First, most of the existing literature is focused on the transportation of containers, without considering freight flow of other cargoes, such as dry bulk and general cargoes, as well as reciprocal impacts between different freight flows to emissions. Pallis et al. (2007) found that over a half of port-related investigations were focused on containers, and the remainder did not specify the cargo type. To study the emissions projection of hinterland transportation, dry bulk should not be neglected as it accounts for up to 60% of the freight volume of port throughput in China (Wang et al., 2014). Furthermore, hinterland freight flows of different types of cargoes do not stand alone, but intertwine with one another, causing reciprocal impacts, thus influencing the emissions of cargoes. For example, an increase in the truck traffic of dry bulk causes highway congestions, thereby extending the transport time of container trucks and causing more freight emissions.

Second, most of the existing literature regards the transport capacity of infrastructures as a constant constraint to freight traffic, and seldom considers the feedback effects of traffic on infrastructures and hence on emissions. The dynamics of infrastructures regarding hinterland transportation in China should be considered. The reasons are twofold. First, dry bulk and some general cargo in China are often transported by large semi-trailer trucks, on which a heavy load, or even an overload that is prevalent in China, may cause significant wear and tear on highway pavement, reducing highway capacity and transport speed and hence increasing emissions. Meanwhile, the Chinese government has exerted tremendous efforts to develop the transportation infrastructure in recent years. As infrastructure construction might take years, it can be neglected for a short-term emissions projection, but must be factored in the long term (de Jong et al., 2013).

Hence, the long-term emissions of hinterland transportation emerge as a result of intertwine and feedback effects within multiple transportation processes, such as freight flow, traffic flow, infrastructure dynamics, and fleet evolution. In this regard, it is necessary to adopt an appropriate approach to address this intertwined hinterland transport process, towards a long-term emissions projection. SD is especially applicable to those problems with nonlinear and feedback features that arise from the intercorrelation of multiple factors in complicated systems (Hjorth and Bagheri, 2006). In this study, an SD model was established to evaluate the long-term effects of alternative environmental policies on hinterland transport emissions.

SD was initially developed by Forrester from MIT in the 1950s and 1960s (Forrester, 1958), and it was first applied to transportation in the 1990s. Abbas and Bell (1994) first confirmed that SD is applicable for transportation modeling, and listed 12 advantages of the approach compared to traditional transport modeling. Shepherd (2014) conducted a comprehensive review on the applications of SD to transportation since the 1990s. In this review, applications of SD to intercity, regional, and national transportation systems were reported. For example, Han and Hayashi (2008) for an SD model of the inter-city passenger transportation in China; Fiorello et al. (2010) for an SD model of the transportation system at the European level; and Egilmez and Tatari (2012) for an SD model of the highway transportation system of the U.S. However, the literature regarding hinterland transportation is still scarce. Recently, Hou and Geerlings (2016) proposed an SD framework to evaluate the effectiveness of alternative port and hinterland operations on sustainability. Unlike this research, that focused more on the macro aspects of the port-and-hinterland system, such as regional economy, transportation, and sustainability. Our study

focuses on specific freight transport processes and transport emissions from the port to hinterland, and vice versa.

This research extends the literature by introducing the SD approach for a comprehensive long-term emissions projection of hinterland transportation. The developed SD model considers some unique characteristics of the hinterland transport processes, such as diversity of cargoes, infrastructure dynamics, and reciprocal influence of freight flows in both hinterland-to-port (H2P) and port-to-hinterland (P2H) directions, which influence emissions directly and indirectly. The results may provide a better understanding of how alternative environmental policies affect long-term emissions through a systematic intervention on a series of hinterland transport processes.

3. Methodology

3.1. Model hypotheses

The objective of this study is to develop an SD model to evaluate the long-term effects of alternative environmental policies on hinterland transport emissions. Based on the objective, the following hypotheses are adopted.

(1) Because a vast majority of emissions are generated in the transportation process, the model considers primarily the transportation that occurs between the port and hinterland, regardless of the emissions caused by freight loading/unloading and handling/stockpiling at both the port and freight owners' yards. The model boundary is hereby illustrated in Fig. 1.
(2) It is assumed that freight owners at the hinterland are gathered primarily in a concentrated area, for example within a city, where they can choose alternative competitive modes. Three typical freight types are considered: dry bulk, containers, and general cargoes, with the corresponding transport modes such as highways (regular highways and express highways are further differentiated in China's context) and railways. Inland waterways, which are not as typical as the aforementioned modes in China, are not considered.
(3) Hinterland transport infrastructures will secure the annual maintenance. Once infrastructure maintenance is conducted, it is assumed that maintenance operations can be completed within a year.
(4) The model assumes that the highway transport costs are only related to the truck payloads, regardless of the impacts of vehicle aging on the constituent transport costs, such as fuel cost, and vehicle maintenance cost.

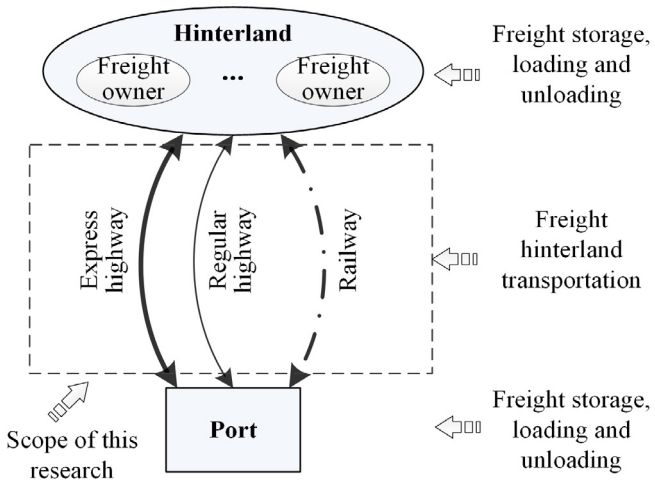


Fig. 1. Illustration of model boundary.

3.2. Model framework

The SD model is developed in a modular manner, consisting of core subsystems developed based on the classic four-step model of transportation (de Jong et al., 2004), and several additional subsystems that are used to address hinterland transportation characteristics. Fig. 2 shows the basic building blocks of the modeling exercise and depicts the feedback mechanism between constituent subsystems.

The logic behind the SD model is that hinterland transport emissions originate primarily from the fuel consumption of freight vehicle traffic that depends on modal traffic volume and transport time, both of which are further affected by a series of dynamically interrelated factors, such as freight volume, fleet configuration, transport capacity of infrastructure, and transport speed and time, all of which constitute the feedback loops. For example, the modal share of a certain period determines the modal freight volume, thus affecting modal traffic volume and transport time, as well as the modal share in the next period. Furthermore, modal traffic causes wear and tear to infrastructures, leading to a dynamic change in the transport capacity of infrastructures, thus affecting the transport time, modal share, and modal traffic volume in the next period.

3.3. Model development

3.3.1. Freight generation and distribution (FGD) subsystem

Freight demand triggers hinterland transportation processes, in which modal traffic flows serve to fulfill the freight demand and generate emissions simultaneously. This subsystem first determines the annual freight volume of each kind of cargo, based on the exogenous prediction of the growth rate of freight volume, and subsequently generates the modal freight volume, according to the modal split ratios provided by the modal split of freight (MSF) subsystem. The generic representation of this subsystem is shown in Fig. 3.

3.3.2. Modal fleet evolution (MFE) subsystem

The modal freight flow comprises freight vehicles traffic, such as diesel and liquefied natural gas (LNG) truck traffic on highways, and diesel and electric train traffic on railways. The fuel consumption of freight vehicle traffic serves as a direct source of carbon and AP emissions. The generic representation of this subsystem is shown in Fig. 4.

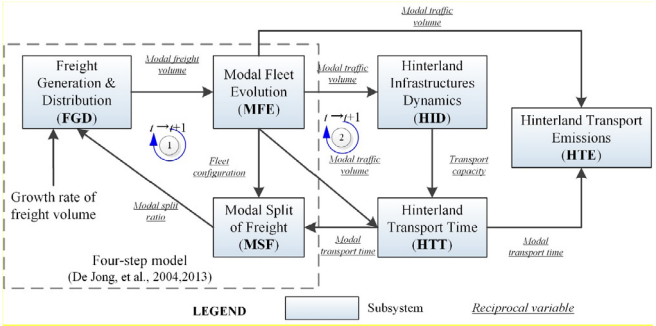


Fig. 2. Structural components of the SD model.

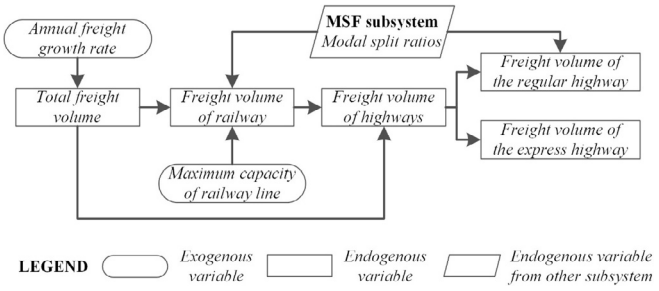


Fig. 3. Generic representation of the FGD subsystem.

This subsystem is dedicated to the following two functions:

(1) Modal traffic calculation. This subsystem considers both the loaded vehicle traffic and empty vehicle traffic, as hinterland transportation consists both H2P and P2H directions and loaded vehicles traversing in one direction may return empty following the other direction. Empty vehicle traffic not only generates a different level of fuel consumption, compared to loaded vehicle traffic, but also shares the infrastructure capacity and affects the transport time of loaded vehicle traffic. This subsystem first considers whether the freight in both directions for a certain transport mode is the same; if it is the same, the subsystem matches them and determines the vehicle type according to the freight type, and subsequently calculates the loaded vehicle traffic volume by the larger freight volume in both directions, and the empty vehicle traffic volume by the difference in freight volume.

(2) Fleet evolution simulation. The modal fleet evolution is simulated based on the classic scrap-and-sales model that calculates annual numbers of new and scrapped vehicles (Huo and Wang, 2012). In addition, considering the significant wear and tear effects of different truck payloads on the loss of truck service life, this subsystem introduces a Weibull distribution function to provide different survival probabilities of trucks with different payloads (Oguchi and Fuse, 2015). Regarding railway vehicles, a fixed survival probability is applied. The railway fleet evolution functions as a sort of conveyor where freight trains are transferred year by year from an age class to the subsequent one and are dropped out after the maximum service life.

3.3.3. Hinterland infrastructures dynamics (HID) subsystem

Freight traffic causes wear and tear to hinterland infrastructures, thereby increasing transport time and emissions owing to a reduction in the transport capacity of infrastructures, while infrastructure maintenance can restore transport capacity. Furthermore, wear and tear on infrastructures caused by loaded vehicles is different from that by empty vehicles. This subsystem simulates the wear and tear and maintenance dynamics of hinterland infrastructures, and estimates the transport capacity of infrastructures accordingly. The generic representation of this subsystem is shown in Fig. 5.

For highway infrastructures, the average transport capacity of highway k in year t, denoted $q_k(t)$, is calculated by the following equation:

$$q_k(t) = i_k(t) \times Q_k \tag{1}$$

where Q_k is the maximum capacity of highway k, regulated by the specifications published by the Ministry of Transport of China (2006); $i_k(t)$ is the average value of the pavement performance indicator (PPI) regarding highway k, obtained by the following empirical PPI deterioration function (Hang and Li, 2010):

$$\begin{cases} i_k(t) = \dfrac{i_k^*(t) + \hat{i}_k(t)}{2} \\ \hat{i}_k(t) = i_k^*(t) \times \left(1 - e^{-a_k(t)^{\beta_k(t)}}\right) \end{cases} \tag{2}$$

where $\hat{i}_k(t)$ is the year-end PPI value, reflecting the pavement condition after the wear and tear of truck traffic, $i_k^*(t)$ is the PPI value at the beginning of the year recovered by pavement maintenance operations (including do nothing (DN), corrective maintenance (CM), and restorative maintenance (RM)), which are determined by $\hat{i}_k(t-1)$. $a_k(t)$ and $\beta_k(t)$ are related to the actual service life of highway k, calculated by the maximum equivalent axles that a highway can bear, and the equivalent axles of truck traffic within a year provided by the MFE subsystem.

For railway lines, it is assumed that the railway facilities will secure the annual regular maintenance because there is no additional damage to line facilities from overloaded freight trains, which are strictly forbidden due to their fatal damage to railway lines as well as railway freight safety. Therefore, the transport capacity of railway lines is maintained at the maximum value, which is specified by the China Railway Corporation (2006).

3.3.4. Hinterland transport time (HTT) subsystem

This subsystem is used to calculate the average modal transport speed and time. The modal transport time not only affects the time cost for freight owners, thus affecting their modal choices (Fiorello et al., 2010), but also influences the fuel consumption and emissions of modal traffic. The generic representation of this subsystem is shown in Fig. 6.

Regarding highway transport speed and time, it is necessary to

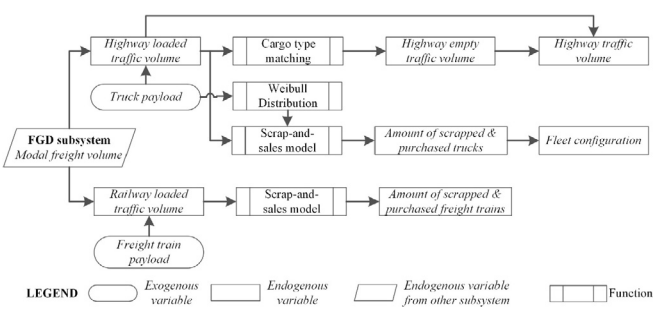


Fig. 4. Generic representation of the MFE subsystem.

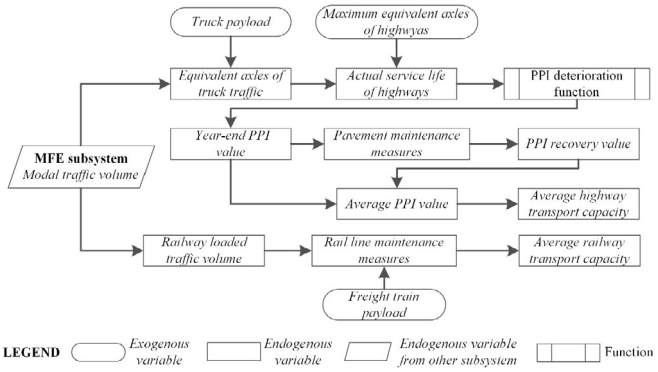


Fig. 5. Generic representation of the HID subsystem.

consider the diversity of truck types owing to multiple types of cargoes. In China, containers and dry bulk are generally transported by large semi-trailer trucks, and general cargoes by smaller truck types (Hang and Li, 2010). This subsystem first converts different types of trucks into unified passenger car units (PCU) and determines the total highway traffic volume (Mardani Nokandeh et al., 2016), and subsequently estimates the average transport speed of trucks on highway k (denoted $s_k(t)$) using a speed-flow function (Saha et al., 2017; Du et al., 2018), as shown in Eq. (3):

$$\begin{cases} s_k(t) = \dfrac{\lambda_k \times S_k}{1 + (v_k(t)/q_k(t))^{\tau(t)}} \\ \tau(t) = w_k + \mu_k \times (v_k(t)/q_k(t))^3 \end{cases} \quad (3)$$

where S_k is the regulated highest speed of highway k, $v_k(t)$ and $q_k(t)$ are the traffic flow and the average transport capacity of highway k, respectively, provided by the MFE and HID subsystem, and the parameters λ_k, w_k, and μ_k are defined by the specifications published by the Ministry of Transport of China (2006).

Regarding freight trains, the average speed depends primarily on technical characteristics of railway infrastructures and locomotives, such as grade of rail lines, traction power, locomotive types, and locomotive traction tons. After determining the average modal transport speed, the average transport time can be obtained easily based on the transport distances.

3.3.5. Modal split of freight (MSF) subsystem

This subsystem aims to calculate the modal shares that are subsequently used by the FGD subsystem to generate the modal freight volumes. The generic representation of this subsystem is shown in Fig. 7.

The modal share is determined primarily using the modal general costs, including transport and time costs (Fiorello et al., 2010). Highway transport cost is calculated by an empirical trucking cost function (Hang and Li, 2010), while railway transport cost is obtained by referring to the price of railway freight service provided by the China Railway Corporation as a third-party transportation. The time cost represents the transport time, fed by the HTT subsystem, multiplied by a time cost factor that is related to the cargo type. The higher the cargo value, the greater is the time cost factor. After the modal general costs are obtained, a nested-logit model is employed to calculate the modal shares for railways and highways, and furthermore, for regular and express highways (Koppelman and Bhat, 2006).

3.3.6. Hinterland transport emissions (HTE) subsystem

This subsystem serves to calculate the carbon and AP emissions, both of which are further unified into the emissions costs, for a comprehensive evaluation of the actual impacts of the total emissions on the environment. The general method of estimating transport emissions is to multiply an emission factor per unit of activity by

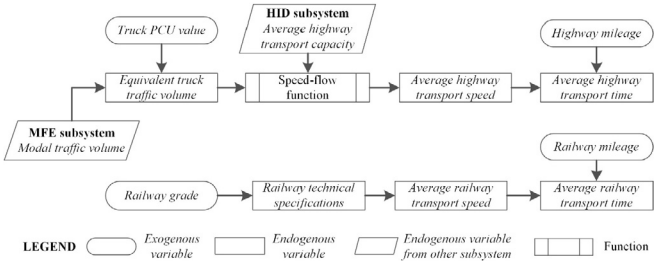


Fig. 6. Generic representation of the HTT subsystem.

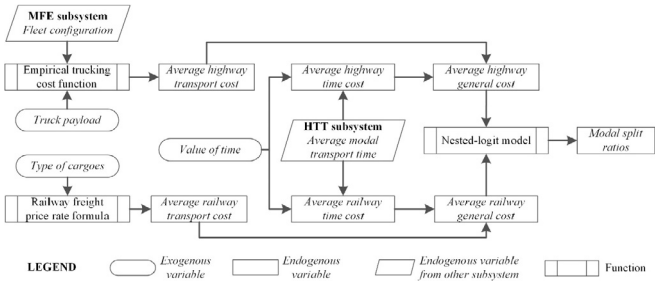


Fig. 7. Generic representation of the MSF subsystem.

the amount of transport activity, including two specific approaches: the fuel-based approach and the distance-based approach (Ubeda et al., 2011). Furthermore, two differentiated levels of aggregation exist, including (1) disaggregated approaches that consider the specific kinematics of the vehicle fleet such as speed and acceleration; (2) aggregated approaches, by which calculations are based on the average speed, annual mileage, and emission factors. This subsystem adopts the aggregated approaches because the objective of this research is a long-term emissions projection. The generic representation of this subsystem is shown in Fig. 8.

The carbon emissions and emissions cost are calculated by the aggregated fuel-based approach. The fuel consumption of modal traffic is first calculated by the modal traffic volume and the average modal transport time, fed by the MFE and HTT subsystems, separately, as well as two correction factors relevant to the vehicle payloads and long-term improvement of fuel efficiency (Fiorello et al., 2010). The carbon emissions of modal traffic are subsequently calculated, according to the carbon emission factors of alternative fuel types, and the carbon emissions cost obtained by the carbon trading price. This study focuses on the emissions directly from modal traffic, implying that railway transportation emissions directly from fuel consumption of diesel locomotives are considered, regardless of those from the electricity consumption of electric locomotives.

The AP emissions and emissions costs are calculated by the aggregated distance-based approach, provided that AP emission factors, regulated by the Chinese vehicle emission standards, are distance based. Four major types of APs are considered: CO, NOx, HC, and PM. First, modal AP emissions are estimated, according to transport mileage, transport time of modal traffic, and emission factors of alternative types of APs. Next, AP emissions costs are obtained, based on the AP emission volume and marginal costs of alternative types of APs.

4. Case study

4.1. Case profile and scenario settings

The developed SD model was applied to a case in China: the

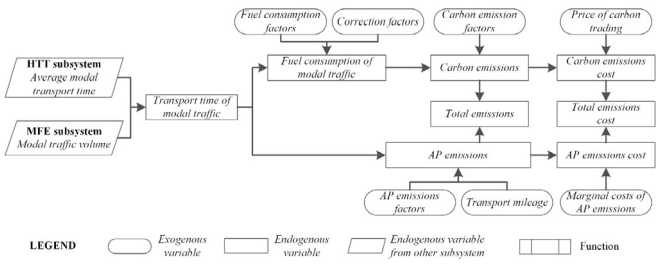


Fig. 8. Generic representation of the HTE subsystem.

hinterland transportation of the Caofeidian port, as illustrated in Fig. 9.

The Caofeidian port is a large port in northern China that primarily handles iron ore and steel. Iron ores imported from overseas are unloaded in the Caofeidian port and transported through highways to Tangshan, a hinterland city that is the largest steel base in China. Steel produced by local steel enterprises in Tangshan is transported to the Caofeidian port and subsequently shipped to southern China. Currently, highways remain the only modal option for both P2H and H2P directions, including a regular highway and an expressway. Particularly for the regular highway, as truck weight inspections are not performed, severe overload trucking exists for both iron ore and steel, causing significant damages to the pavement and traffic safety issues. The primary truck type is the six-axle semi-trailer diesel truck, with a van trailer for iron ore and flatbed trailer for steel, both of which cannot be used interchangeably. Therefore, both freight trucks return empty. To eliminate severe overloading of trucks on the regular highway and mitigate the potentially increasing emissions caused by the continuously growing throughput of the Caofeidian port, the local government intends to enforce several environmental policies that are formulated as a combination of basic measures and additional measures, as shown in Table 1.

Based on the case profile and scenarios settings, some of the key parameters of the SD model are shown in Table 2.

Simulations were performed after the model parameters were set. The simulation period encompasses 23 years beginning from 2008, when the Caofeidian port was first put into operation, to 2030, the final year of China's fifteenth five-year plan. The simulations performed on historical years serve as model validation, and those on future years as policy scenario projection.

4.2. Model validation

The proposed model was validated following the process of SD model validation suggested by Barlas (1996), including behavior reproduction and sensitivity tests. Behavior reproduction tests are exemplified by two variables: pavement maintenance measures and average highway transport time, as their physical meanings are explicit and the historical data are easy to obtain. The results, as shown in Fig. 10, have demonstrated that the simulated values are well correlated with the actual historical values. This validation process is also conducted on other variables, and the results are satisfactory.

The behavior sensitivity is tested, as exemplified by the freight volume of the express highway on the P2H direction. In the test, the highway transport cost per unit freight volume is adjusted to a 20% fluctuation around its current value. The results (see Fig. 11) indicate that the highway freight volume is sensitive to the change in transport cost, which are consistent with the actual circumstances.



Fig. 9. Illustration of the case study.

4.3. Results and analyses

1. The scenario comparisons on modal shares are shown in Fig. 12.

As the basic measure of all the optional policies, the RWR counteracts the scale economies of transportation as it reduces the truck payload significantly, thus resulting in the increased trucking cost of the regular highway and rendering the regular highway less competitive compared to the express highway. Consequently, the modal share of the regular highway on both the P2H and the H2P directions decreased significantly (see the line marked 2 in Fig. 12). Regarding the RPL scenario, the modal shares are almost the same as that of the RWR scenario, because although LNG trucks cost less per unit volume than diesel trucks, the competitiveness of both highways barely changes when the diesel trucks on both highways are replaced simultaneously by LNG trucks (see the line marked 3 in Fig. 12). Regarding the RPM scenarios, the modal shares vary along with the railway grade (see the lines marked 4, 5, and 6 in Fig. 12). The higher the railway grade, the greater is the freight volume shifted by the railway, and the less are the freight volume and traffic flow on highways. Consequently, the advantage of the express highway on speed and time cost is weakened; hence, a higher modal share for the regular highway is obtained.

2. The scenario comparisons on modal traffic are shown in Fig. 13.

As shown, although the modal share of the regular highway is reduced under different policy scenarios (see Fig. 12), the traffic volume still increases, because the RWR reduces the truck payload significantly, resulting in a substantial increase in the total truck traffic volume. Only under the RPMI scenario owing to the sufficient transport capacity of the grade-I railway, can the traffic flow be approximately equal to the baseline.

3. The scenario comparisons on highway maintenance measures are shown in Fig. 14. A cumulative maintenance value is shown, with different maintenance measures assigned with different values (specifically, DN, CM, and RM equals to 0, 0.5, and 1, respectively), to reflect a cumulative effect of alternative policies on highway pavement conditions.

As shown, regarding both the RWR and RPL scenarios, as the RWR reduces the truck payload of the regular highway significantly, the pavement of the regular highway is protected and maintenance measures are shifted from RM to CM, leading to a lower cumulative maintenance value, as is shown in Fig. 14a. However, as freight is shifted from the regular highway to the express highway (see modal shares shown in Fig. 12), an increasing traffic flow on the express highway causes a surging RM and a higher cumulative maintenance value, as shown in Fig. 14b. Regarding the RPM scenarios, the railway diverts the freight of highways, thus protecting the pavements of both highways.

4. The scenario comparisons on the highway transport time in the P2H direction are shown in Fig. 15.

Regarding the RWR and RPL scenarios, the transport time of the regular highway is slightly shorter than that of the baseline (see Fig. 15a) because the effect of an enhanced transport capacity of the regular highway, caused by pavement protection (see Fig. 14a), contributes more than that of an increased traffic volume (see Fig. 13a); meanwhile, the transport time of the express highway is remarkably higher than that of the baseline (see Fig. 15b), owing to

Table 1
Proposed policies and corresponding scenarios.

Scenario name	Basic measures	Additional measures	Scenario profile
Baseline	None	None	The current situation continues, with no environmental policies implemented.
RWR (rigid weight regulation)	RWR	None	From 2018, trucks payload is not allowed to exceed the statutory limit.
RPL (RWR plus LTR)	RWR	LTR (LNG trucks replacement)	From 2018, in addition to the enforcement of RWR, scrapped diesel trucks are replaced by LNG trucks.
RPMIII/II/I (RWR plus MSRIII/II/I)	RWR	MSRIII/II/I (Modal shift from highways towards railway of grade III/II/I)	From 2018, in addition to the enforcement of RWR, constructing a local railway in two years, with potential grade III/II/I, respectively, to shift iron ore away from highways.

Table 2
Parameters and values of the SD model.

Parameters	Initial value	Units
Freight volume of iron ore (P2H)	9.37×10^6	ton/year
Freight volume of steel (H2P)	3.50×10^6	ton/year
PCU of the trucks on the express highway	7.0	PCU/vehicle
PCU of the trucks on the regular highway	5.5	PCU/vehicle
Mileage of the express highway	90.7	km
Mileage of the regular highway	94.9	km
Mileage of the local railway	110	km
Payload of trucks on the express highway	35	ton/vehicle
Payload of trucks on the regular highway	93	ton/vehicle
Statutory limit of truck payload	35	ton/vehicle
Traction ton of the railway locomotive (grade III/II/I)	2500/4000/5000	ton/train
Maximum capacity of the local railway (grade III/II/I)	$10 \times 10^6/20 \times 10^6/30 \times 10^6$	ton/year
Value of time for iron ore	70	Yuan/hour
Value of time for steel	350	Yuan/hour
Carbon emission factor of diesel/LNG	3.16/2.83	ton/ton
AP emission factors of diesel (CO/NOx/HC/PM)	$2.1 \times 10^{-6}/5 \times 10^{-6}/0.66 \times 10^{-6}/0.1 \times 10^{-6}$	ton/(hour*km)
AP emission factors of LNG (CO/NOx/HC/PM)	$0.21 \times 10^{-6}/3.05 \times 10^{-6}/0.19 \times 10^{-6}/0$	ton/(hour*km)

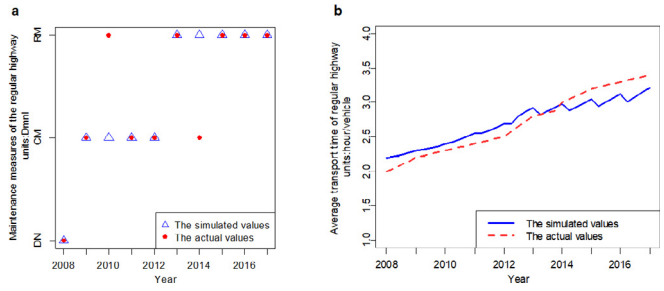


Fig. 10. Behavior reproduction tests on (a) the maintenance measures and (b) the average transport time of the regular highway.

both an increasing modal traffic volume and a decreasing transport capacity (see Fig. 14b). Regarding the RPM scenarios, because both the traffic volume and pavement wear and tear of highways are reduced by a freight shift from the highways to the railway, the transport time of both highways is reduced significantly; the higher the railway grade, the shorter is the highway transport time.

The scenario comparisons on the highway transport time in the H2P direction are shown in Fig. 16.

As shown, for the RWR and RPL scenarios, the transport time of both highways increases significantly. Given that the loaded traffic volume of the H2P direction is much lower (see freight volumes in Table 2), the results are primarily from the empty truck traffic returning from the P2H direction, in which the loaded traffic volume is much higher. Therefore, although the railway in the RPM scenarios is not meant to divert steel in the H2P direction, it reduces the volume of the empty truck traffic returning from the P2H direction, and hence shortens the highway transport time in the H2P direction.

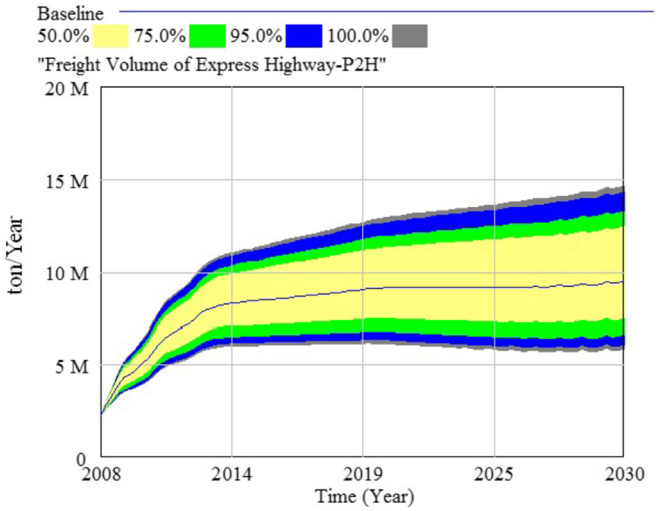


Fig. 11. Behavior sensitivity test on the freight volume of the express highway.

5. The scenario comparisons on hinterland transport emissions are obtained. Fig. 17a and b shows the carbon and AP emissions stock by year, respectively, and Fig. 18 shows the total emissions stock volume at the end of the simulation period.

The results indicate the following:

(1) Regarding the RWR scenario, as the freight volume increases with time, the influences of reduced truck payload to the increase in both truck traffic volume (see Fig. 13) and transport time (see Figs. 15 and 16) become increasingly evident.

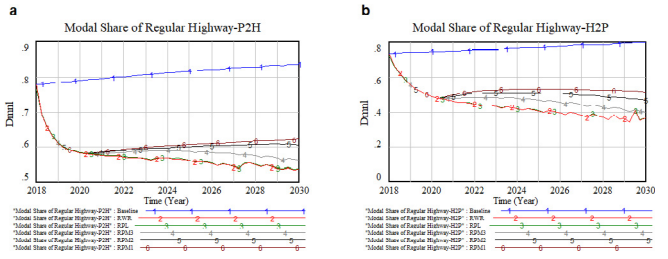


Fig. 12. Modal share of the regular highway in the (a) P2H direction and (b) H2P direction.

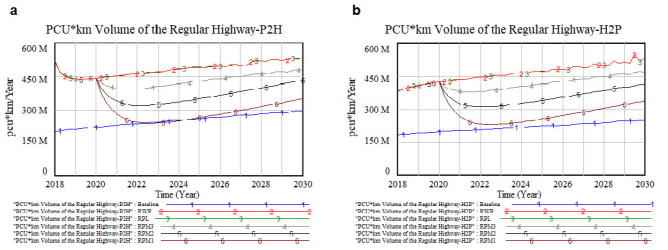


Fig. 13. Traffic volume of the regular highway in the (a) P2H direction and (b) H2P direction.

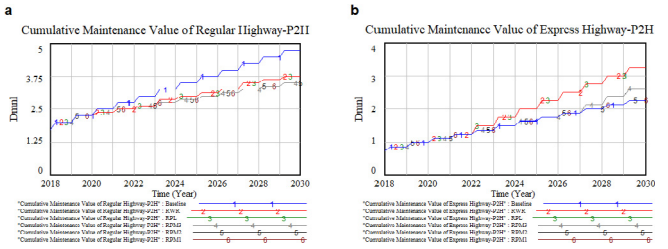


Fig. 14. Cumulative maintenance value of (a) the regular highway and (b) the express highway.

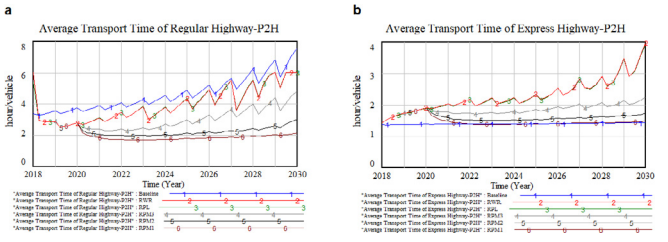


Fig. 15. Transport time of (a) the regular highway and (b) the express highway in the P2H direction.

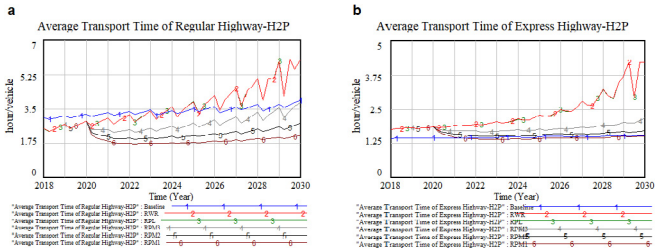


Fig. 16. Transport time of (a) the regular highway and (b) the express highway in the H2P direction.

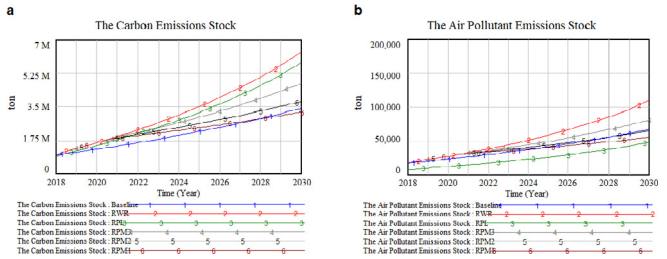


Fig. 17. Scenario comparisons on (a) the carbon emissions stock and (b) the AP emissions stock.

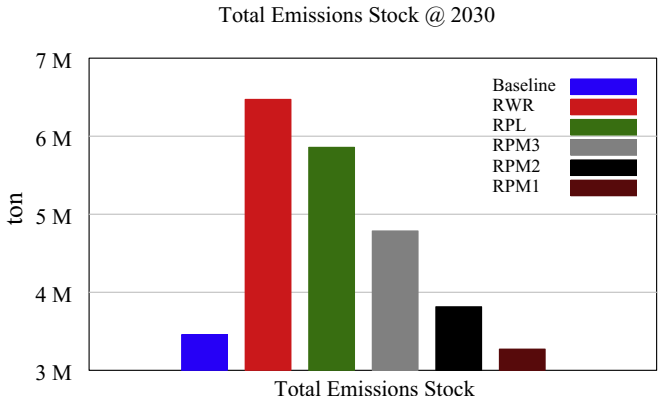


Fig. 18. Total emissions stock at the end of the simulation period.

Consequently, both the carbon and the AP emissions stock in the RWR scenario are the highest, as shown in Fig. 17a and b, respectively.

(2) Regarding the RPL scenario, under which the modal shares, modal traffic volume, and transport time are the same as those of the RWR scenario, the carbon emissions stock is slightly lower than that of the RWR scenario, because of the slightly smaller carbon emission factor of LNG compared to that of diesel, and the AP emissions stock is significantly lower than all the other scenarios owing to the remarkably smaller AP emissions factors of LNG compared to those of diesel, as shown in Fig. 17a and b, respectively.

(3) Regarding the RPM scenarios, the results indicate that the scenario with a higher grade of railway yields a lower emissions stock, as illustrated in Fig. 17. This is because the emissions per unit freight volume of diesel trains is lower than that of highway trucks, owing to the scale economies of transportation. The railway with higher grade shifts more freight away from the highways, leading to a fewer emissions stock. However, when the truck payload exceeds its statutory limit significantly, similar to the baseline situation, the scale economies of the highways are enhanced, thus yielding reduced overall emissions. Additionally, highway trucks in the RPM scenarios are still under the payload regulation of the RWR. Consequently, only the total emissions stock in the RPMI scenario, in which the railway with the largest transport capacity is operated, is lower than that of the baseline, as shown in Fig. 18.

In addition to the emissions volume, the emissions costs are further calculated based on the marginal costs of different types of emissions, as shown in Fig. 19 and Fig. 20.

As shown, at the end of the simulation period, the costs of the total emissions stock in both the RPMII and the RPMI scenarios are

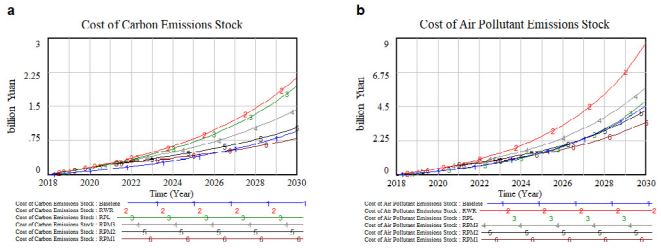


Fig. 19. Scenario comparisons on the costs of (a) carbon and (b) AP emissions stock.

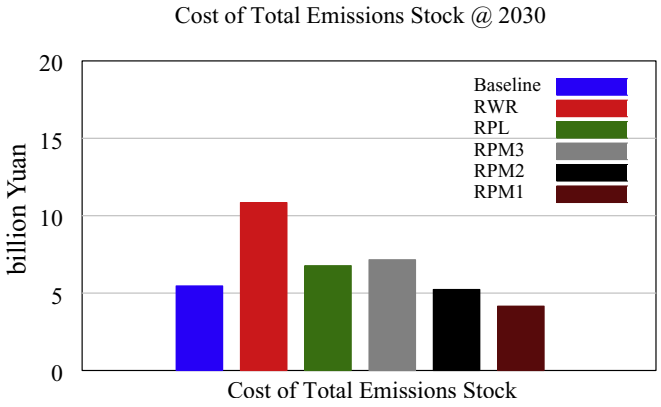


Fig. 20. Cost of total emissions stock at the end of the simulation period.

lower than that of the baseline. This is because the effects of AP emission reduction in the RPM scenarios are strengthened when the marginal costs of the emissions are considered, as the marginal costs of AP emissions are much higher than that of carbon emissions.

5. Discussion

The results indicate that the baseline scenario, in which the trucks are overloaded, yields a high performance in reducing emissions. In contrast, the RWR scenario, in which the truck weight is regulated under the statutory limit, generates the highest volume of emissions. This is because overload trucking enhances the scale economies of transportation. Particularly for the aforementioned case in which the primary cargo is dry bulk, which features a high freight volume and a low cargo value, overloading of trucks exhibits the advantages of reduced truck traffic and transport time, thus reducing emissions significantly. Hence, this conclusion is consistent with that of Hang and Li (2010), who reported that the moderate weight regulation, by which the truck weight is allowed to exceed the statutory limit to a moderate degree, is the most appropriate option to reduce transport and pavement maintenance costs.

Despite the advantages of reducing transport cost and emissions, overloaded trucks are prohibited by the Chinese government as they damage highway pavement significantly and endanger traffic safety. Hence, modal shift is an effective method to protect highway pavements, increase traffic safety, and reduce hinterland transport emissions simultaneously. Hou and Geerlings (2016) reported that modal shift measures (shifting freight from highways to inland waterways) demonstrate a higher potential of reducing CO_2 emissions of hinterland transportation than other measures (e.g., technological and spatial measures). The results of this study also indicate that a modal shift to railways generates fewer emissions than the baseline. However, owing to the high freight volume and truck overload rate (over 150%), the railway must reach grade I to achieve a lower emissions stock than the baseline; this translates to higher government investment than those for grade-II and grade-III railways.

The SD model explicitly illustrates how environmental policies affect long-term emissions of hinterland transportation through a systematic intervention on a series of intercorrelated transport processes. In this regard, decision-makers should focus on not only issues that affect hinterland transport emissions directly, such as cargo type, freight volume, and traffic volume, but also issues related indirectly to emissions. For example, (i) the capacities of transport infrastructures that can be reduced by wear and tear from vehicle traffic and recovered by maintenance, thus affecting transport time and hence transport emissions; (ii) the freight of both P2H and H2P directions, as the vehicle traffic of one direction can be affected by the returning vehicle traffic of the other direction, causing reciprocal consequences of congestion and increased transport time and emissions.

6. Conclusions

The long-term emissions projection of hinterland transportation must consider the dynamical and feedback effects within the hinterland transport processes, such as freight flow, traffic flow, infrastructure dynamics, and transport time. This study contributes to the literature by developing an SD model for a comprehensive long-term emissions projection of hinterland transportation. The SD model was validated and applied to a specific case in China. The results indicate that if the emissions stock volume was considered, the RPMI policy, namely the rigid truck weight regulation and the construction of a grade-I railway, would be effective in emissions mitigation. Furthermore, if the cost of emissions stock was considered, both RPMI and RPMII would be effective. Through a comprehensive comparison of different policy scenarios, the SD model demonstrated explicitly how alternative environmental policies would affect long-term emissions through a systematic intervention on a series of hinterland transport processes.

One limitation of this study is that the SD model did not consider the implementation cost of those formulated policies, e.g., the cost of replacement of diesel trucks by LNG trucks, and the construction cost of the local railway. Therefore, if all the additional measures mentioned in Table 1 are implemented simultaneously, the emissions stock would be reduced further, but the policy implementation cost would be higher. Hence, further work should address both the long-term emissions of hinterland transportation and the implementation cost of emission mitigation policies, to provide a cost-optimal solution if multiple feasible policies exist for emission mitigation.

Declarations of interest

None.

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